

YEAR 10 SCIENCE
PHYSICAL SCIENCES
INVESTIGATION - ACCELERATION DUE TO GRAVITY

AIM To determine the acceleration due to gravity by videoing a falling object.

BACKGROUND

One of the equations used to describe motion in Physics is:

$$s = ut + \frac{1}{2}at^2$$

In this investigation, the object will fall from rest, so $u = 0$ m/s.

If $u = 0$, then the equation becomes:

$$s = \frac{1}{2}at^2$$

Rearranging this equation in terms of acceleration (a) becomes:

$$a = \frac{2s}{t^2}$$

By using the camera on your smart phone or iPad, you can find how far the object falls (displacement - s) in a time (t).

METHOD

1. Go to the settings of your device and check the frame-rate of the camera.
Write it down in the results section.

e.g. iPhone 7s - slo-mo records at 240 frames/second (default setting).
2. Place the object (block of wood, ball, etc.) against a scale attached to a wall.

e.g. 30 cm rule, metre rule, paper scale in centimetres, scale on whiteboard.
3. Record the fall of the object.
4. Go to the file in "photos" and select it. Select "edit".
5. Scroll the video and count how many frames the object moves for a given distance.

e.g. displacement (s) = 5.5 cm = 0.055 m (Convert to metres.)

time = 20 frames

6. Determine the time by:

$$t = \frac{\text{number of frames}}{\text{frame rate}}$$

$$\text{e.g. } t = \frac{20}{240} = 0.0833 \text{ s}$$

7. Calculate the acceleration.

$$\text{e.g. } a = \frac{2s}{t^2} = \frac{2 \times 0.055}{(0.0833)^2} = 15.8 \text{ m/s}^2$$

RESULTS

Frame rate: _____

Displacement (s): _____ cm = _____ m

Time (t): _____ frames = _____ = _____ s

Calculation: $a = \frac{2s}{t^2}$
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CONCLUSION:

The acceleration due to gravity is _____ m/s².